9/28/2025 Briana-Yves 1-1 Meeting Minutes

Problem Statement

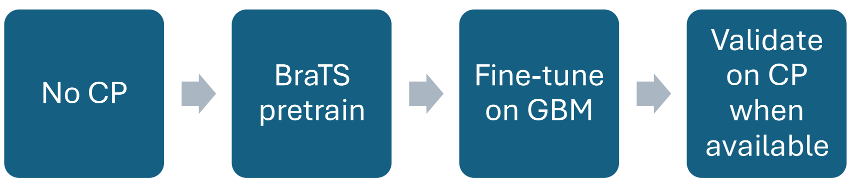
We don't have access to the CP images.

Proposed workaround pipeline

We discussed several workaround strategies:

1. Train a segmentation model on a curated BraTS 2017 dataset.
2. Then fine-tune it on available data:

   - 60 Glioblastoma (GBM) cases
   - A complete dataset of multifocal GBM cases, with validation performed through the disconnectomics tools or the lesion quantification toolkit (LQT).
   - Once available, validate the model on the 90 CP scans.



Tool Description

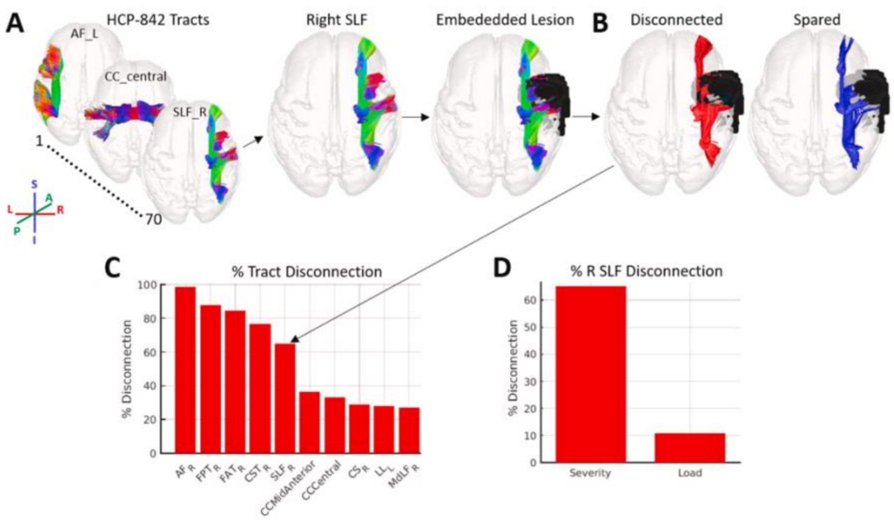
We also discussed different tools that could be evaluated in relation to MRI scans:

- Neuroimaging-based lesion analyses, such as voxel-based lesion-symptom mapping (VLSM, [1]), which show a lesion's spatial localization.

- Atlas-based approaches, which measure a lesion's impact on gray matter regions and white matter connections, as well as the disruption of white matter pathways. One such publicly available tool is the MATLAB Lesion Quantification Toolkit [2].

- Another related analysis is a supervised machine learning method, the Support Vector Regression-based Lesion Symptom Mapping (SVR-LSM), which models the relationship between lesion data and behavioral scores [3].

- Other techniques include disconnection mapping (DSM, [4]) or connectome lesion-symptom mapping (CLSM, [5])



White matter disconnection severity tracts identified by LQT

References

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Tool	Input	Output	Pros	Cons
VLSM	Lesion masks + behavioral scores	Lesion–symptom localization	Well-established; anatomy–behavior link	Needs large number of scans; sensitive to registration
LQT (MATLAB)	MRI lesion masks; atlas	Estimates region-level grey matter damage and white matter disconnection	Publicly available	Atlas accuracy
SVR-LSM	Lesion masks + behavior	ML model	Flexible; handles high-dimensional data	Needs behavioral data
DSM	Lesion masks + tract atlas	White matter disconnection maps	Highlights network disruptions	Requires diffusion atlas
CLSM	Lesion masks + connectome	Network lesion mapping	Network-level insight; connectomics	Needs high-quality connectome

Comparison of Lesion Analysis and Mapping Tools